



Priyanshu

Bachelor of Technology
in Computer Science and Engineering
Indian Institute Of Technology, Ropar

+91-8837819138
2022csb1107@iitrpr.ac.in
GitHub | LinkedIn

EDUCATION

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Indian Institute of Technology, Ropar	7.59 (Till 8th Sem)	2022-2026
Senior Secondary	CBSE – GMSSS-MHC, Chandigarh	85.2%	2021
Secondary	CBSE – GMSSS-MHC, Chandigarh	79.8%	2019

EXPERIENCE

- Full-Stack Web Development intern** May 2025 – Jul 2025
VK-Global **Certificate**
 - Developed a full-stack educational portal using Next.js, React, Node.js, and MongoDB for students, teachers, and admins.
 - Implemented features like real-time quizzes, discussion forums, screen time analytics, and content management.
 - Built secure multi-role authentication with JWT, OTP verification, and role-based access control. AWS S3, Socket.IO, and responsive design for scalable, real-time performance

PROJECTS

- NS-3 Based Emergency Evacuation Network Simulator** Aug 2025 – May 2026
B.Tech Project under Dr. Jagpreet Singh **GitHub | Demo**
 - * Simulated an IoT fire evacuation system in NS-3 with 100 Wi-Fi sensors and Dijkstra-based real-time path computation.
 - * Diagnosed PDR degradation (24%→98%) via ECMP routing fix; evolved polling to a reliable event-driven UDP protocol.
 - * Implemented hybrid emulation using TapBridge with a real C++ Base Station on the host OS.
 - * Built a React/Konva web topology editor for visual NS-3 configuration, deployed on Vercel.
- Safarnama – Travel Companion App** Jan 2025 – May 2025
Development Engineering Project, by Dr. Puneet Goyal **GitHub**
 - * Developed an Android travel app enabling students to plan and document group trips with a modern UI.
 - * Designed and implemented the backend using Node.js, Express.js, and MongoDB, building APIs and data models from scratch and implemented features like collaborative trips, travel blogs, fuzzy search, notifications etc.
 - * Deployed on Google Cloud VM with Nginx setup and PM2 for process management; integrated GitHub Actions for CI/CD.
- ChampSim Simulator Optimization** Mar 2024 – Apr 2024
Computer Architecture Project **GitHub**
 - * Tackled high cache conflict misses in CPUs by implementing and optimizing new cache indexing and replacement strategies.
 - * Utilized ChampSim simulator to rigorously test, compare, and automate trace analysis for cache management algorithms.
 - * Achieved a measurable reduction in cache misses, significantly improving overall system efficiency.
- Minor Projects**
 - * RISC-V Assembler – C++ compiler converting assembly to machine code, live register and memory states. **GitHub**
 - * Branch Predictor – C++ based predictor to enhance instruction flow efficiency. **GitHub**
 - * Portfolio Website – Personal portfolio with interactive UI, project showcase, and AI chatbot integration. **GitHub**
 - * AIMS Portal Clone – Full-stack web app with OTP-based login, course enrollment, and grade assignment. **GitHub**

TECHNICAL SKILLS

- **Languages:** C/C++, Python, Java, RISC-V Assembly
- **Tools/Frameworks:** HTML, CSS, JavaScript, Node.js, MongoDB, SQL, Arduino, NS-3

KEY COURSES TAKEN

- **CSE:** Algorithms & Data Structures, Operating Systems, DBMS, Digital Logic, Computer Architecture, Algorithms Design, Computer Networks, Blockchain, Software Engineering, Theory of Computation, Blockchain and Cryptocurrencie, Wireless Adhoc Networks.
- **Mathematics::** Calculus, Linear Algebra, Differential Equations, Probability and Statistics
- **Others::** English Language Skills, Signals and Systems, Electronics, Economics, Ethics, Bio-MEMS

MISCELLANEOUS

- **JEE Advanced**, Secured AIR around 13k 2022
- **IISER Aptitude Test**, AIR around 5k 2022
- **ISMP Mentor**, 2024